9TH JANUARY 2025/26

deti
universidade de aveiro
departamento de eletrónica, telecomunicações e informática

Speech and Gesture Fusion for Multimodal Excel Interaction

Multimodal Interaction

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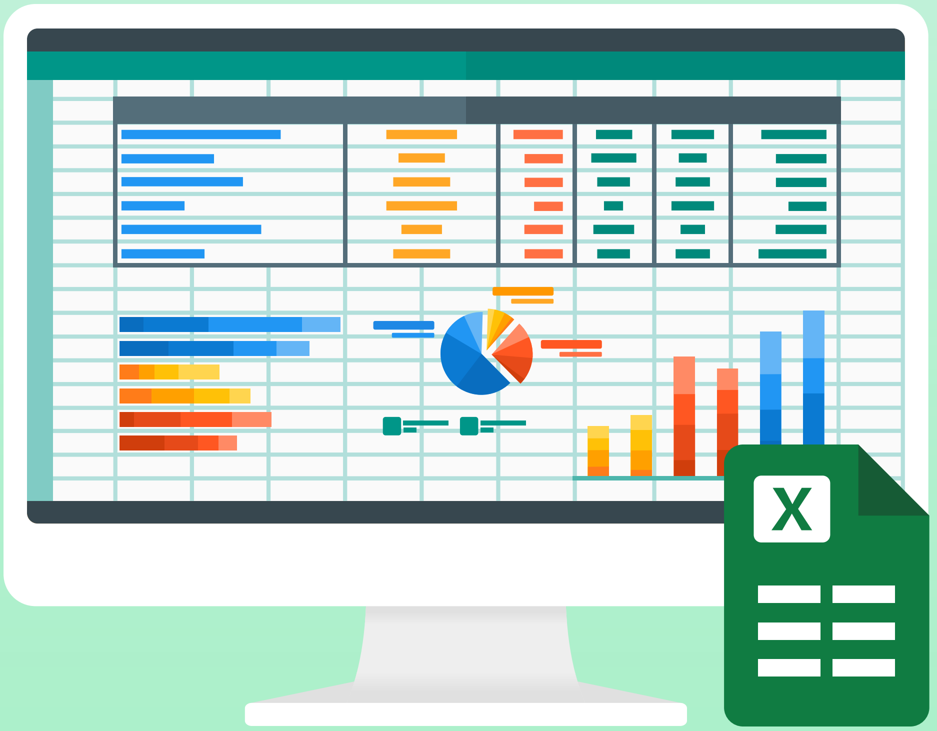

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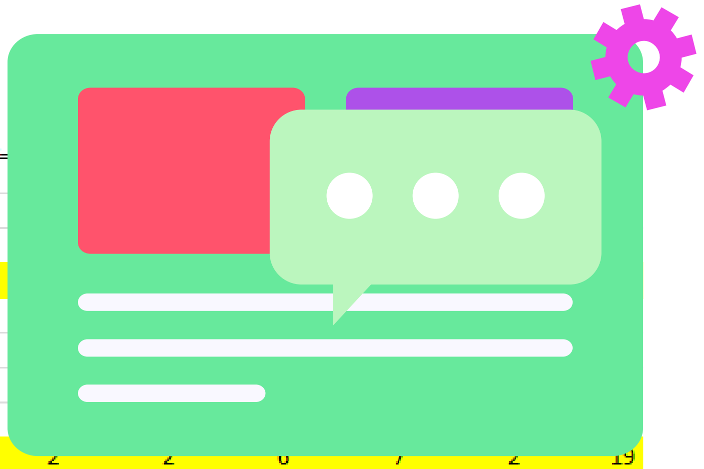
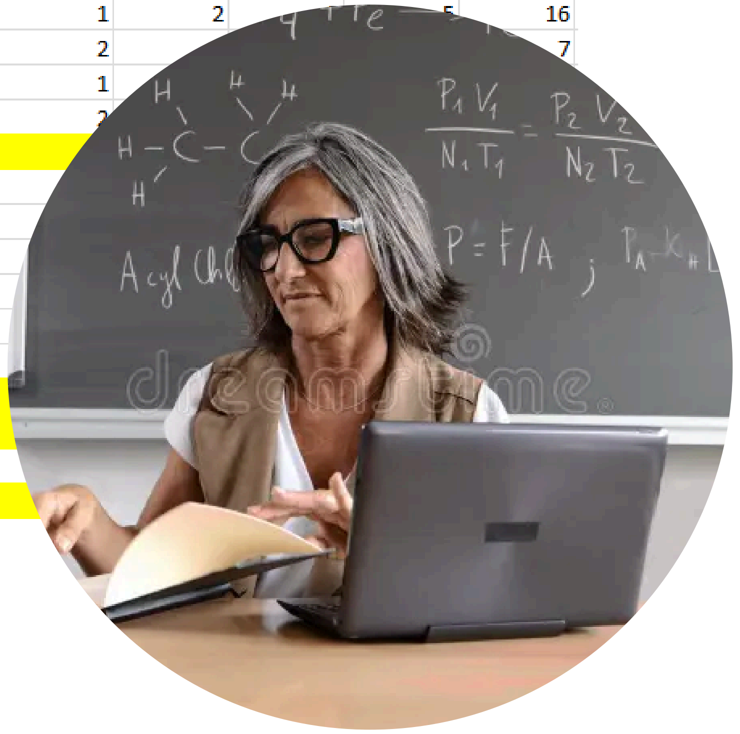
01 Introduction

Persona

- **Name**: Prof. Helena Martins
- **Age**: 58 years old
- **Field**: Social Sciences / Education
- **Experience**: 30 years of university teaching
- **Digital Skills**: Intermediate (uses email, Moodle, and PowerPoint, but avoids advanced tools)

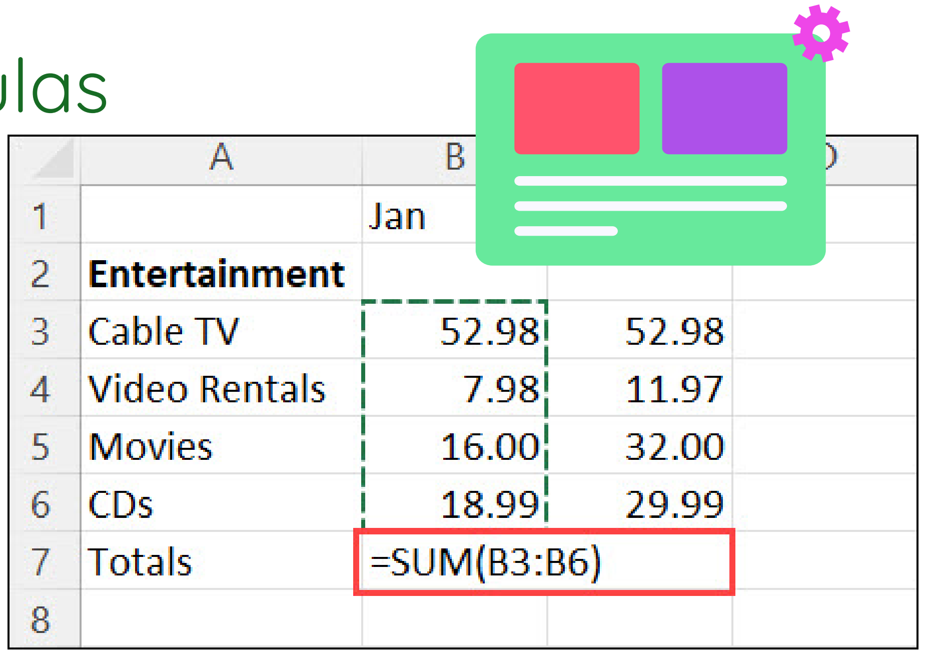
mecano	Nome	ereço de e-	digo de cu	TP	P	REGIME	Teste 1	T2_					
107447	MARIA INÊ	maria.ines	8314	TP2	P1a	O	13						
94716	ANA CATA	acatarinaf	8246	TP1	P8	O	8						
107135	MARIANA	marianam	8314	TP2	P2	O	11						
107390	MARIA JO	mariaforte	8314	NO_TP			11						
108121	DANIELA I	dani.teles	8314	TP1	P10	O	15						
107242	JOANA M(	joanamms	8314	TP1	P7	O	17						
108589	JOANA GC	joana.inac	8314	TP2	P6	O	11						
108522	JOSÉ PEDR	jpaabreu@	8315	TP1	P7	O	13						
107218	EDUARDA	eduardaca	8314	TP2	P5	O	10	2	2	6	7	2	19
108189	JOSÉ PEDR	gasparbar	8314	TP2	P2	O	12	1	1	2	5	4	13
107372	ELIANA PE	eliana.bet	8314	TP2	P2	O	15	6	4	1	4	1	16
108957	DIOGO GA	diogo.gcar	8314	TP1	P8	O	14	1	5	2	5	5	18
107440	PEDRO GA	gab.olivei	8314	TP2	P1	O	16	5	1	2	5	2	15
108188	JOÃO MAI	joaomat22	8314	TP2	P1	O	16	1	2	1	2	1	7
108855	MARIANA	marianagn	8314	TP1	P8	O	18	2	1	7	1	4	15
108672	MARIANA	marianaco	8314	TP1	P8	O	9	1	4	1	1	5	12
108138	DIOGO MI	diogo.orfa	8314	TP2	P1	O	14	3	1	2			16
107236	BEATRIZ M	beatriz.afc	8314	TP1	P8	O	15	1	2				7
104204	ALEXANDE	alexander	8307	TP1	P8	O	15	6	1				
107428	INÊS MAR(	i.melo@u	8314	TP1	P10	O	16	1					
107259	DANIEL FIL	daniel.mo	8314	TP1	P10	O	9	2					
100088	MARIA MI	maria.cast	8307	TP2	P1	O	13	6					
108506	CAROLINA	carolinaca	8314	TP2	P1	O	17	1					
110376	RICARDO (	rfigueired	8314	TP2	P1	O	16	1					
109038	MARTA BC	martabdg(	8314	TP1	P10	O	17	5					
108727	JOÃO PEDI	joaorolo03	8314	TP1	P10	O	12	1					
93973	CATARINA	cdmartins	8307	TP1	P10	O	11	3					
108381	MARIANA	marianale	8314	TP1	P10	O	8	3					
108694	MIGUEL XA	miguelxlir	8314	TP1	P10	O	11	1					
108421	AFONSO R	marques.a	8314	TP2	P1	O	13	1					
108366	LEONOR M	leonor.ma	8314	TP1	P10	O	8	2					
109462	JOÃO HEN	joaohenri	8314	TP1	P10	O	10	5					





01 Introduction
Persona

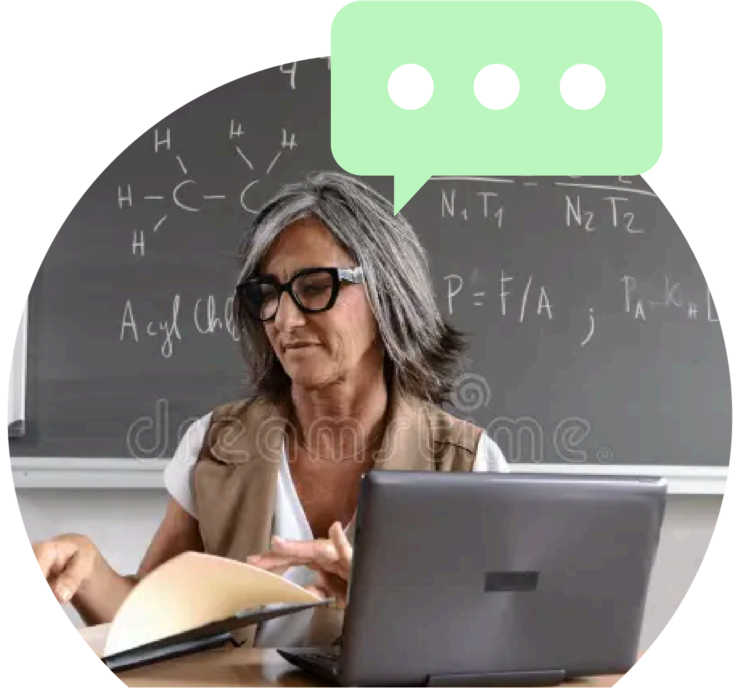
- **Difficulty using advanced Excel features**
  - Difficulty calculating averages
  - Difficulty applying conditional formatting
  - Difficulty generating charts
  - Difficulty creating pivot tables
  - Particularly struggles with writing and understanding formulas
- **Fear of causing errors in the spreadsheet**
  - She feels insecure when working in Excel, mainly because she is afraid of accidentally deleting important data or altering the layout or structure of the sheet,



	A	B		D
1		Jan		
2	**Entertainment**			
3	Cable TV	52.98	52.98	
4	Video Rentals	7.98	11.97	
5	Movies	16.00	32.00	
6	CDs	18.99	29.99	
7	Totals	=SUM(B3:B6)		
8				

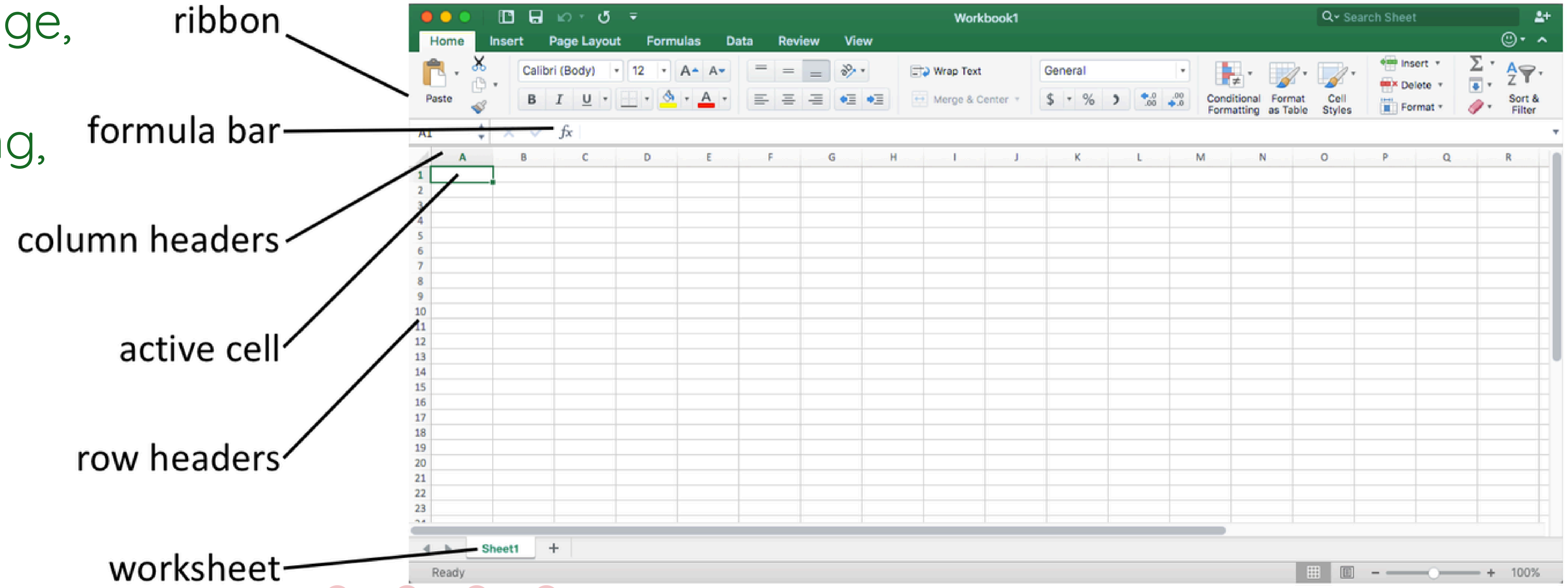
01 Introduction

Persona



- **Difficulty understanding Excel’s technical vocabulary**

  Common Excel terminology is not intuitive for her, including:
  - column, row, cell, range,
  - pivot table,
  - conditional formatting,
  - bar chart.



01 Introduction

Goals

1. **Automate common Excel tasks using multimodal interaction**
    - Use speech and gestures together to perform Excel actions naturally.
2. **Improve task execution efficiency through multimodal control**
    - Complete repetitive Excel tasks faster by combining voice and gesture input.
3. **Support hands-free interaction**
    - Control Excel without mouse or keyboard.
4. **Increase accessibility through multimodal commands**
    - Make Excel easier to use for less experienced users.
5. **Reduce user errors through multimodal fusion and confirmation**
    - Prevent mistakes through controlled multimodal interaction.

01 Introduction

Scenario

Context

- Professor Helena Martins is preparing the class final grades in Excel. While most grades are straightforward, some students require individual performance analysis before the final results are submitted.

Interaction

- While working from home, she feels more comfortable speaking aloud and combining voice commands with gestures, but still prefers a simple and reliable way to interact with Excel due to limited confidence with advanced features..
- By combining different interaction modalities, the system reduces cognitive load, minimizes errors, and improves efficiency

Outcome

- The gesture-based interaction allows her to work silently, reduce errors, and complete the task more efficiently and confidently.

01 Changes Since Previous Assignments

Assignment 1

- **Existing command logic was improved**
- **Confirmation mechanism** was added for a critical action
- **Fallback** voice command was enhanced.
- **Several commands** were enhanced, including custom columns, pivot tables, and cell value updates.

Assignment 2

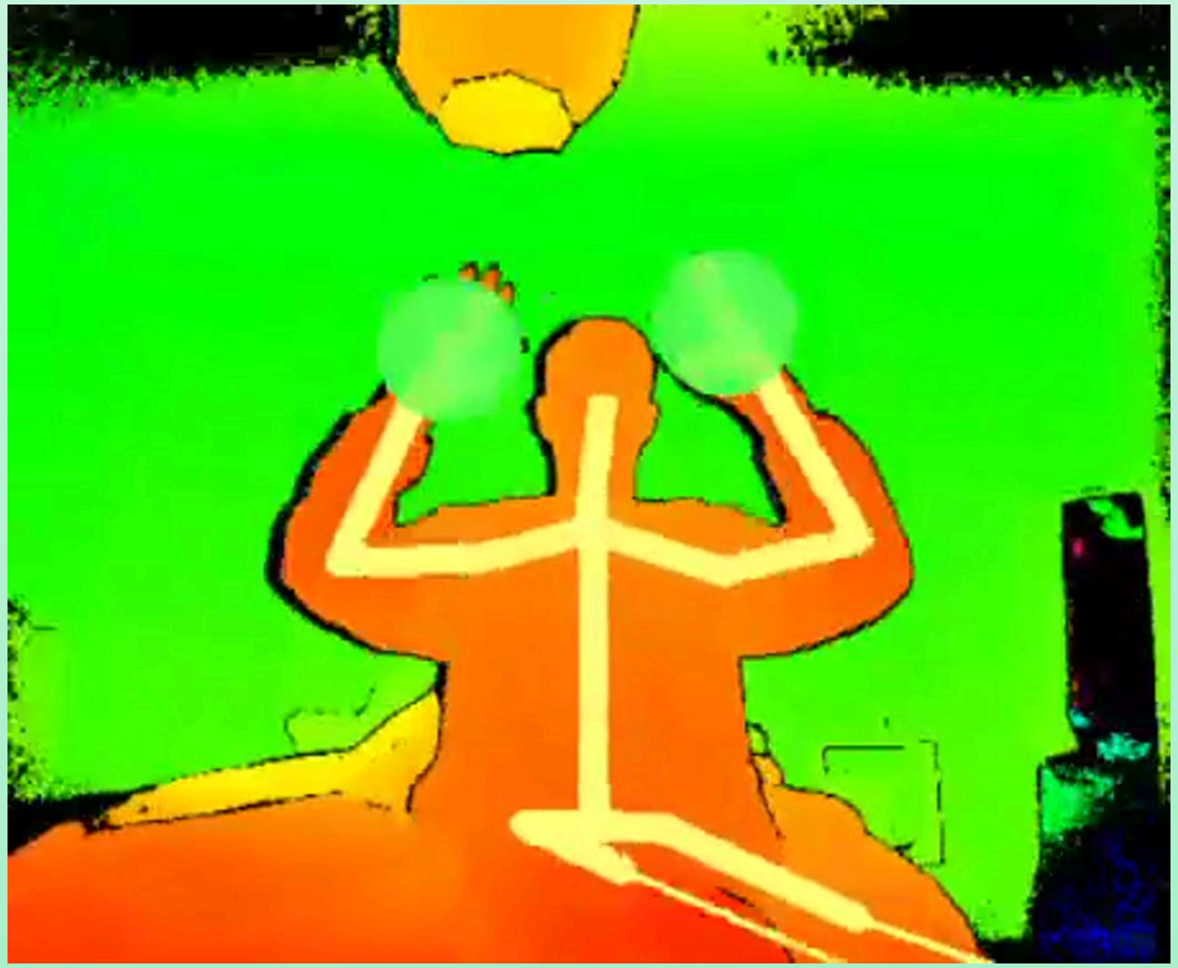
- Added a confirmation mechanism for **Undo Last Action**, requiring the gesture to be repeated within a short time window
- Changed the **Close Excel command** to a more intuitive home gesture and changed **the Student Approved and Student Failed commands.**
- **Re-recorded and significantly improved existing gestures**
- Improved gesture recognition performance, achieving an average confidence of **0.55**
- Added **speech output to the system**
- **Enabled communication between the web application and the backend to provide vocal feedback**
- Added a new gesture **Hand Grab**

01 Changes Since Previous Assignments

Implemented Intents - Gestures

Explanation: Closes the application quickly and safely

Close Excel



Close the Excel application

01 Implemented Intents - Gestures

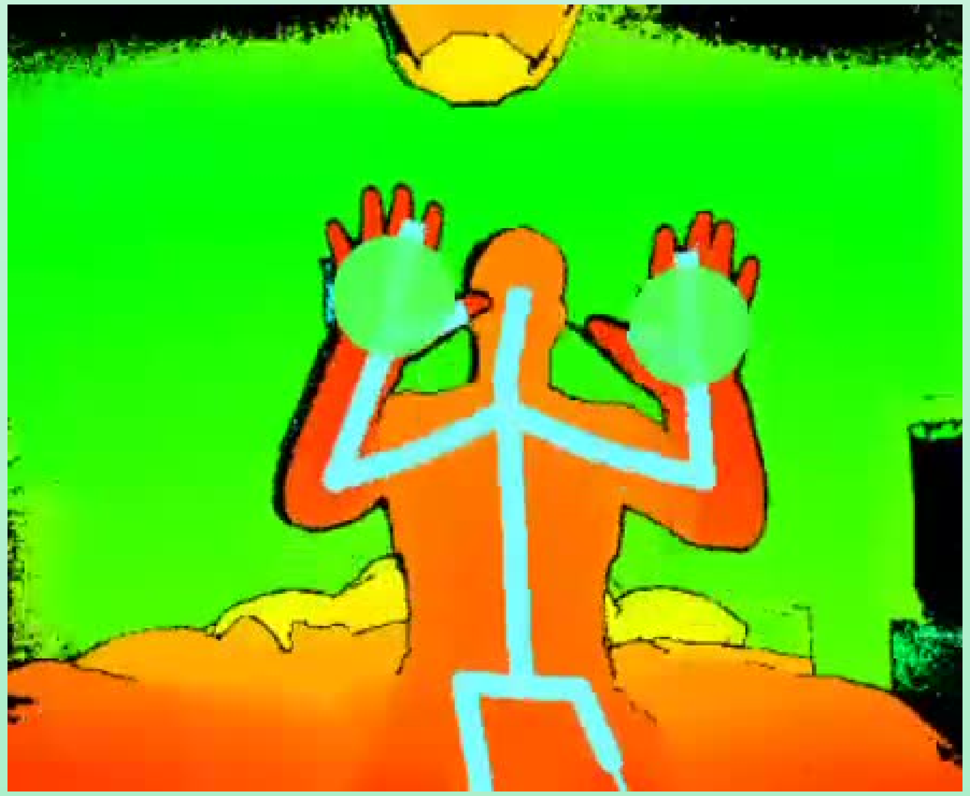
Implemented Intents - Gestures

Explanation: Supports visual analysis of results

Highlight Failed

Excel Action
Highlight falling values

Highlight Passed



Excel Action
Highlight passing values

02 System Architecture Overview

User

Fusion Engine (MMI Framework)

Combines speech and gesture events
Implements redundancy and complementarity
Generates high-level commands

Excel

Microsoft Excel

Controlled via Excel Interop
Executes spreadsheet actions
Provides visual feedback to the user

Interaction Manager (IM)

Receives events from all modalities
Routes events to Fusion or Application

Input Layer

Speech Modality (Voice Commands)
Gesture Modality (Kinect / Touch)

Application Layer

Backend (C#)

User → Speech & Gesture Modalities → Interaction Manager → Fusion Engine → Backend (C#) → Microsoft Excel

03 Implemented Intents - Gestures

Excel Voice Commands vs. Gesture Commands

Voice Commands

- Calculate average
- Highlight passed and failed students
- Insert columns
- Apply real improvement
- Apply possible improvement
- Perform mathematical operations
- Insert questions
- Generate class performance charts
- Generate student bar charts
- Generate question-based charts
- Delete all charts
- Save file / report
- Update grades
- Create pivot table
- Display help
- Confirmation
- Cancel

Gestures Commands

- Zoom in
- Zoom out
- Navigate up
- Navigate down
- Navigate left
- Navigate right
- Calculate average
- Insert column
- Highlight passed students
- Highlight failed students
- Clear cell content
- Close Excel
- Select cell / Select range

03 Implemented Intents

Redundancy, Complementarity and Single Modality

Single-Modality Interactions

(same action, voice OR gesture)

- Zoom in (gesture)
- Generate class performance charts(voice)
- Generate student bar charts(voice)
- Generate question-based charts(voice)
- Delete all charts(voice)
- Zoom out (gesture)
- Navigate up (gesture)
- Navigate down (gesture)
- Navigate left (gesture)
- Navigate right (gesture)
- Highlight results(voice)

Approved + failed students

Redundant Interactions

(same action, voice AND gesture)

- Calculate average (voice / gesture)
- Insert column (voice / gesture)
- Highlight passed students (voice / gesture)
- Highlight failed students (voice / gesture)
- Close Excel(voice/ gesture)

03 Implemented Intents

Redundancy, Complementarity and Single Modality

Complementary Interactions

(modalities must be combined)

- Select cells **(Hand Grab + navigation gestures)** → Calculate average **(voice)**
- Select a range **(gesture)** → Generate a chart **(voice)**
- Select a range **(gesture)** → Generate a bar chart **(voice)**
- Select students **(gesture)** → Highlight results **(voice)**
- Insert a column **(voice)** → Highlight passed students **(gesture)**
- Insert a column **(voice)** → Highlight failed students **(gesture)**
- Undo the last action **(gesture)** → Confirm action **(voice)**
- Close Excel **(gesture)** → Confirm action **(voice)**
- Delete All Chart **(gesture)** →Confirm action **(voice)**

04 DEMO

04 DEMO

Demo Scenario – Professor Workflow

1. Select a student row
2. Calculate the average grade for the selected student
3. Insert a status column for the student (Approved / Failed)
4. Highlight the student based on approval status
5. Generate student bar charts
6. Navigate through the spreadsheet
7. Operations
8. Undo the last action after a mistake
9. Close Excel

05 Challenges & Conclusion

Challenges

- **Multimodal recognition accuracy**

  Ensuring reliable recognition of both gestures and voice commands, while avoiding false positives and missed inputs.
- **Modality differentiation and complementarity**

  Clear and intuitive coordination between gestures and voice.
- **User variability across modalities**

  Handling differences in users' body sizes, movement styles, speaking pace, accents, and pronunciation.
- **Real-time multimodal responsiveness**

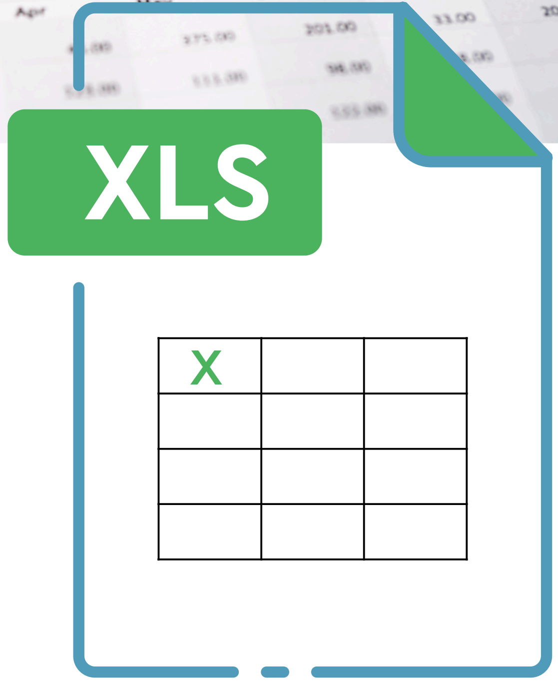
  Low latency between multimodal input and Excel actions.
- **Integration of multimodal components**

  Seamless coordination of gestures, voice, fusion, and the Excel backend.

05 Challenges & Conclusion

Conclusion

- Multimodal interaction makes **Excel more intuitive and accessible.**
- It reduces dependence on **menus and complex commands** by combining gestures and voice.
- The system successfully executes Excel actions using coordinated gestures and voice commands.
- Future work includes expanding the **multimodal command set and improving gesture-voice integration.**



THANK YOU!

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